

Trace-Based Comparative Evaluation of FreeRTOS and ZephyrRTOS in an Energy Meter

Real Time Operating Systems 25ES611

PROJECT REPORT

Submitted by

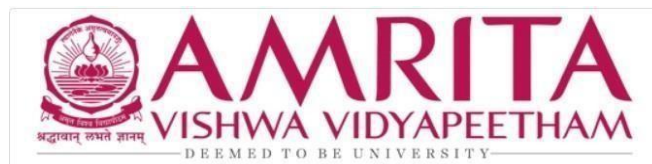
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CHAPTER 1: INTRODUCTION

The field of embedded systems has undergone a fundamental transformation, shifting away from simple, linear "bare-metal" programming toward sophisticated, multi-threaded architectures. This shift has made the choice of a Real-Time Operating System (RTOS) one of the most consequential decisions in the early stages of product design. Two platforms have emerged as leaders in this space: FreeRTOS and the Zephyr Project. FreeRTOS, a long-standing industry standard, is prized for its minimalist, deterministic kernel, low memory overhead, and straightforward API, making it the default choice for millions of resource-constrained microcontroller applications. In contrast, Zephyr, supported by the Linux Foundation, represents a modern, modular, and vendor-neutral approach. It features a robust, device-tree-driven hardware abstraction layer designed specifically to solve the scalability and portability challenges common in massive-scale IoT deployments.

This project investigates the performance trade-offs between these two systems when implemented on the ESP32—a dual-core, Wi-Fi and Bluetooth-enabled microcontroller that sits at the intersection of hobbyist experimentation and industrial edge computing. The motivation for this study is to understand the architectural overhead and developer-experience costs associated with each RTOS by utilizing a practical energy-metering application as a testbed. This system, which monitors a 9W resistive load, necessitates precise timing for ADC sampling, thread safety via mutexes, and I2C peripheral management, creating a perfect environment to evaluate scheduling efficiency and resource contention.

Through this analysis, we aim to uncover how FreeRTOS's lightweight design compares to the structured, safer, yet heavier abstraction provided by Zephyr. By measuring variables such as mutex block time and power consumption stability, we provide a comparative benchmark that assists developers in choosing the optimal RTOS for high-performance IoT projects. This study serves as a practical guide for engineers who must balance the competing requirements of real-time responsiveness, code maintainability, and rapid time-to-market. Ultimately, this report provides a quantitative and qualitative foundation for selecting an RTOS that aligns with the specific hardware constraints and scalability needs of modern intelligent IoT devices.

1.1 Problem Statement

- 1 Modern IoT devices, like smart energy meters, require efficient multitasking to handle high-frequency sensor sampling and user interface updates simultaneously. Choosing between **FreeRTOS** and **Zephyr RTOS** on the ESP32 involves a critical trade-off: FreeRTOS offers a lightweight kernel with minimal overhead, while Zephyr provides a modular, device-tree-based architecture that prioritizes portability and code safety.
- 2 The core problem is the **lack of empirical benchmarking** comparing these two architectures. Developers currently lack objective data on how the "abstraction cost" of Zephyr impacts real-time scheduling latency (mutex blocking time) and system responsiveness compared to the manual, bare-metal-like control of FreeRTOS. This project addresses this by implementing an automated Hardware-in-the-Loop (HIL) framework to quantify these performance differences, providing a data-driven basis for selecting an RTOS in resource-constrained IoT applications.

2.2 Objective

The primary goal of this project is to objectively evaluate the performance and architectural differences between FreeRTOS and Zephyr RTOS when deployed on the ESP32 platform. To achieve this, the following objectives have been defined:

- **Implement a Multi-Tasking Architecture:** Develop a standardized energy-metering application that executes four concurrent tasks—ADC sampling, data calculation, I2C display updates, and system monitoring—on both FreeRTOS and Zephyr.
- **Establish a Hardware-in-the-Loop (HIL) Testbed:** Construct a reliable testing environment using a 9W resistive load to ensure that the ESP32's performance metrics are evaluated under identical, stable, and reproducible conditions.
- **Quantify Real-Time Scheduling Latency:** Measure and compare the "Mutex Block Time" across both OS environments to determine the scheduling overhead imposed by Zephyr's hardware abstraction layer versus the lightweight nature of FreeRTOS.
- **Develop an Automated Benchmarking Framework:** Design and deploy a Python-based automation suite that handles the entire lifecycle of the test—compiling, flashing, and logging performance data—to eliminate human error and variability.
- **Create a Comparative Performance Dashboard:** Visualize the captured metrics (power consumption stability and scheduling latency) in a real-time PyQt6 dashboard to provide a data-driven visual comparison of RTOS performance.
- **Formulate an OS Selection Matrix:** Compile the experimental findings into a technical summary that serves as a guide for developers, highlighting the specific project requirements where the portability of Zephyr outweighs the raw efficiency of FreeRTOS.

CHAPTER 2: LITERATURE SURVEY

Real-Time Operating Systems are widely used in embedded applications where multiple operations must be performed concurrently with predictable timing. In RTOS-based systems, the application can be divided into separate tasks, and each task can be assigned a priority based on its importance. The Percepio RTOS 101 white paper explains that RTOS-based design improves maintainability by dividing software into smaller independent tasks such as control, communication, and human-machine interface functions. It also explains that most RTOSes use fixed-priority scheduling, where the scheduler selects the highest-priority ready task for execution. This concept is directly related to the present project, where voltage/current measurement is assigned higher priority than calculation and display/trace-related operations.

Palatty et al. presented a performance analysis of a FreeRTOS-based video capture system using SEGGER SystemView. Their work highlights the importance of RTOS in applications where multiple functions must be scheduled, synchronized, and managed under timing constraints. The authors explain that a super-loop approach becomes difficult when the system complexity increases, because task priority, synchronization, and shared resource management are harder to handle without an RTOS. They implemented the application functions as FreeRTOS tasks and used SystemView to analyze task behavior, computational requirements, and timing constraints. This is closely related to the present work, where an ESP32 energy meter is divided into FreeRTOS tasks and analyzed using SEGGER SystemView.

The same study also emphasizes the need to analyze execution time, processor utilization, task frequency, context switching, and task dependencies in real-time embedded systems. The authors state that the SystemView profile provides information such as system events, context switches, task periodicity, task priority, execution time, processor utilization, and stack size. This supports the methodology used in the present project, where SEGGER SystemView is used to observe task activations, CPU load, total runtime, blocked time, scheduler activity, and idle time for the FreeRTOS-based energy meter.

Katkoori et al. developed a FreeRTOS-based multisensor environmental monitoring system using an ESP32. Their work shows how FreeRTOS can be used to manage multiple sensor tasks with priority-based scheduling and preemptive multitasking. The system used message queues for synchronized data exchange between tasks and displayed sensor data locally and remotely through an IoT dashboard. The authors observed that FreeRTOS improved task execution time, latency, CPU utilization, and scheduling behavior compared with a non-RTOS implementation. This study supports the use of FreeRTOS in the present energy meter project, where measurement, calculation, and display or trace-related operations are separated into independent tasks.

The multisensor monitoring work also demonstrates that FreeRTOS improves modularity and responsiveness in embedded systems. The paper reports that the FreeRTOS-based system reduced execution time and improved responsiveness by allowing time-sensitive sensor tasks to execute with less delay. It also mentions the use of priority inheritance to avoid priority inversion when shared resources are accessed. This is relevant to the present project because mutexes are used to protect shared variables such as voltage, current, power, and energy. The use of mutual exclusion ensures safe data transfer between measurement and calculation tasks.

Surya et al. proposed “Watt Watcher,” an RTOS-powered energy meter based on ESP32 and FreeRTOS. The system used voltage and current sensors, ADC sampling, RMS computation, display updates, and Blynk cloud communication as independent RTOS tasks. The paper highlights that FreeRTOS features such as task prioritization, core pinning, queues, semaphores, and software timers help achieve reliable real-time energy monitoring. The authors also mention that high-priority ADC sampling tasks were protected from lower-priority operations such as display and Wi-Fi communication. This is directly related to the present project, where the energy meter workload is divided into measurement, calculation, and trace/display-related tasks.

The Watt Watcher paper further validates the importance of RTOS in energy monitoring applications by showing that FreeRTOS improves temporal precision, system response, task reliability, and data integrity. The authors report that FreeRTOS-managed communication was assigned lower priority so that Wi-Fi delays did not affect voltage or current sampling. They also observed that the modular multitasking design allowed OLED refresh, cloud updates, and energy summation to operate without resource contention. This supports the design decision in the present project to give the measurement task the highest priority, since ADC sampling and RMS processing are the most time-critical operations in the energy meter.

Boddula et al. presented an IoT-based pollution monitoring system using Zephyr RTOS and AWS cloud services. The system used ESP32, multiple sensors, cloud communication, and local display. The authors explain that Zephyr RTOS supports real-time task scheduling, multithreading, and priority-based preemptive scheduling. Their work shows that Zephyr can be used to manage sensor acquisition, display updates, and cloud communication in real-time IoT systems. Although Zephyr implementation was not completed in the present project due to toolchain and tracing setup limitations, this paper provides support for the planned future comparison between FreeRTOS and Zephyr RTOS.

The Zephyr-based pollution monitoring study also shows that ESP32 is suitable for IoT and RTOS-based applications because of its processing power, wireless connectivity, and support for communication interfaces such as UART, I2C, SPI, and ADC. The paper describes Zephyr as a scalable and configurable

RTOS suitable for IoT applications requiring multitasking, networking, and real-time processing. This is relevant to the present work because the ESP32 is also used as the processing unit for voltage and current acquisition, power calculation, and runtime tracing.

Overall, the reviewed literature shows that RTOS-based design is useful for embedded monitoring systems because it enables modular task design, priority-based scheduling, synchronization, and improved real-time responsiveness. Previous studies on FreeRTOS-based video processing, multisensor monitoring, and energy monitoring demonstrate the importance of tracing and profiling tools for understanding runtime behavior. The literature also supports the use of SystemView-like trace analysis to study task execution time, CPU load, scheduling behavior, and resource usage. Based on these studies, the present project implements an ESP32-based energy meter using FreeRTOS and analyzes its runtime behavior using SEGGER SystemView.

CHAPTER 3: METHODOLOGY

The experimental methodology employed a Hardware-in-the-Loop (HIL) approach, designed to isolate the performance characteristics of the operating system from the physical hardware variables. By utilizing a fixed 9W LED bulb as a stable resistive load, the test environment provided a constant power draw, ensuring that any observed jitter or performance variance was directly attributable to the RTOS scheduler rather than fluctuating electrical demands. The hardware architecture utilized an ESP32 microcontroller as the central processing unit, with a signal acquisition chain consisting of an ACS712 Hall-effect sensor for current, a precision voltage divider for potential measurement, and an I2C-based 16x2 HD44780 LCD for real-time feedback.

The software was architected into four distinct threads to simulate a realistic IoT workload: a high-priority measurement task, a math-intensive calculation task, an I/O-bound display task, and a background monitoring task. The FreeRTOS implementation, built using the Arduino-ESP32 framework, utilized imperative programming patterns where tasks were spawned at runtime using `xTaskCreatePinnedToCore`. Resource sharing was managed via manually initialized `SemaphoreHandle_t` mutexes, providing high-speed, low-overhead access to the global energy metrics. In contrast, the Zephyr implementation utilized a declarative approach, leveraging Zephyr's Device Tree and configuration files to define hardware mappings. Tasks were defined statically using `K_THREAD_DEFINE`, which optimized memory layout by pre-allocating stack space at compile-time, prioritizing code safety and maintainability over raw execution speed.

To eliminate human bias, a custom Python-based HIL orchestration suite was developed to act as a master controller. This script performed a three-stage sequence: automated deployment, synchronized logging, and data visualization. First, it invoked `esptool.py` for FreeRTOS and the `west` build tool for Zephyr via subprocess commands to automate the flashing process, ensuring the latest binaries were always deployed. Once the device reset, the script opened a UART serial bridge and used RegEx-based parsing to normalize the output streams, mapping "Mutex Block Time" and "Power Watts" into a unified data structure. Finally, the results were processed in real-time by a PyQt6 interface, which overlaid the FreeRTOS and Zephyr traces on synchronized graphs. This methodology transformed the development workstation into a high-precision testing laboratory, enabling reliable side-by-side performance analysis.

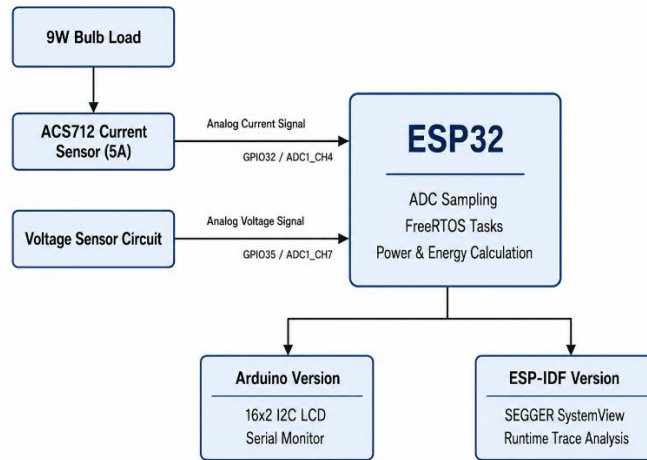


Figure 3.1: Block Diagram

3.1 Hardware Components

The hardware setup for this energy metering system was engineered to ensure high-fidelity data acquisition while maintaining a compact, safe footprint suitable for HIL testing. At the core of the system is the ESP32 microcontroller, which provides the dual-core processing power and high-resolution ADC channels necessary to handle the concurrent sensing and calculation tasks. The system is fed by a 230V AC mains supply, which powers the 9W LED bulb load. To measure this, an ACS712 Hall-effect current sensor is inserted in series with the load, translating the alternating current into a proportional analog voltage. Simultaneously, a precision voltage divider circuit steps down the 230V AC mains to a safe, low-voltage signal compatible with the ESP32's input tolerances. Both signals are fed into the ESP32's 12-bit ADC channels, which are configured with 11dB attenuation to maximize the readable input range and provide precise instantaneous measurements.

The user-interface layer is managed by a 16x2 character LCD, interfaced via an I2C PCF8574 backpack module. This module simplifies the I/O requirements, allowing the display of voltage, current, power, and accumulated energy using only the SDA and SCL lines of the ESP32. By utilizing a common ground between the 230V sensing circuitry and the ESP32's 3.3V logic supply, the system maintains signal integrity, reducing the "off-baseline" noise often seen in high-impedance AC sensing. The entire assembly is powered via a standard USB-to-UART bridge, which provides the necessary 5V power for the LCD and sensor suite while simultaneously enabling the high-speed UART stream used by the Python benchmarking orchestrator. This integrated setup creates a stable, real-world load environment that allows for rigorous, repeatable performance testing across the FreeRTOS and Zephyr RTOS platforms.

3.2 FreeRTOS Task Architecture

To optimize real-time performance and prevent resource contention, the FreeRTOS architecture is structured into four distinct tasks, each assigned a specific priority to ensure that critical sampling routines are never starved by background processes. The task priority system follows a preemptive model, where higher numerical values denote higher importance to the scheduler.

- **MeasTask (Priority 3):** As the most critical component of the energy meter, this task is dedicated to high-frequency ADC sampling of the current and voltage sensors. By assigning it the highest priority, we ensure that the timing of the sampling window remains jitter-free, which is essential for accurate RMS calculations. This task is pinned to Core 1 to maximize execution throughput.
- **CalcTask (Priority 2):** This task handles the intensive floating-point mathematics required to convert raw ADC samples into RMS voltage, current, and cumulative Watt-hour energy values. It operates at a lower priority than the measurement task, ensuring that calculations only occur once new data is available without interrupting the ongoing sampling process.
- **LCDTask (Priority 1):** Responsible for the user interface, this task manages the I2C communication required to update the 16x2 display. It is assigned the lowest priority because the physical update rate of the LCD (once per second) is significantly slower than the high-frequency measurement loop. This prevents the slow, blocking I2C protocol from interfering with real-time sensor data.
- **MonitorTask (Priority 1):** Operating at the same priority as the LCD task, this thread is an observer that collects system telemetry, such as free heap memory and individual task status. It runs at a lower frequency (every five seconds) to avoid consuming unnecessary CPU cycles, serving primarily to feed data to the external benchmarking dashboard via the UART serial link.

This prioritization ensures that the measurement and calculation tasks—the "brain" and "eyes" of your energy meter—remain highly responsive, while the display and monitoring tasks serve as lower-priority background processes. This design keeps the system stable even when the display is updating or the serial monitor is active, preventing any data loss during the metering process.

3.4 ZephyrRTOS Architecture

The Zephyr RTOS architecture follows a strictly declarative model, where task scheduling and hardware access are managed through the kernel's configuration and device tree systems rather than imperative code. In this project, the Zephyr implementation utilizes a static thread definition approach, which pre-allocates resources at compile-time to ensure deterministic behavior. Similar to the FreeRTOS setup, the task priorities are configured to ensure that high-frequency data acquisition is protected from background interference, though Zephyr's preemptive scheduler handles these priorities through a highly integrated, kernel-managed lifecycle.

- **MeasThread (Priority 3):** This thread is the system's primary data acquisition engine, responsible for continuous ADC polling of the current and voltage sensors. Because Zephyr's thread priorities are mapped to kernel-level scheduling, this thread is assigned a higher priority to ensure that the sampling frequency remains constant. By utilizing Zephyr's internal interrupt service routines, this thread achieves high-precision timing, ensuring that RMS calculations are derived from jitter-free datasets.
- **CalcThread (Priority 2):** Serving as the computational heart of the system, this thread consumes the raw sensor data to perform RMS and power integration. It is assigned a medium priority, allowing it to yield to the measurement thread during active sampling bursts while ensuring that the calculation interval remains synchronized with the meter's 1-second reporting cycle.
- **LCDThread (Priority 1):** This thread handles I2C communication with the PCF8574-backed LCD. In Zephyr, this thread is decoupled from the physical I2C hardware by the kernel's driver abstraction

layer. It is assigned a lower priority, ensuring that even if the I2C bus experiences latency or blocking wait-states, the higher-priority measurement and calculation threads continue to function without interruption.

- **MonitorThread (Priority 1):** Similar to the FreeRTOS implementation, this background thread manages system health and telemetry. By enabling `CONFIG_THREAD_ANALYZER=y`, this thread automatically collects stack usage and CPU cycle data from the kernel's scheduler. It runs at the lowest priority, periodically waking up to stream the system status over the UART bridge for the external benchmark orchestrator.

Unlike the dynamic allocation pattern used in FreeRTOS, Zephyr's architecture relies on the Device Tree to map these threads to the underlying hardware, providing a more robust, albeit structured, approach to task execution. Each thread in Zephyr is part of a managed execution environment where the kernel enforces strict resource bounds, ensuring that no single thread can exceed its allocated stack space or interfere with the global kernel state. This approach provides higher reliability for long-term industrial deployments, as the static allocation prevents the memory fragmentation that can sometimes occur in dynamic, long-running systems.

CHAPTER 4: RESULT AND DISCUSSION

- The experimental deployment of the energy metering firmware utilizing the Zephyr RTOS provided measurable insights into its scheduling efficiency, memory management, and system stability under continuous sensing loads. Utilizing the kernel's internal thread_analyzer module, system telemetry was captured during steady-state 230V, 9W RMS load conditions to evaluate real-time performance.
- **Table 1: Zephyr RTOS Performance Metrics**

Thread / Metric	Priority	CPU Load (%)	Peak Stack Usage (Bytes)
Measurement (meas_tid)	3 (High)	37%	464 / 4096
Calculation (calc_tid)	4	0%	304 / 4096
LCD Display (lcd_tid)	5	0%	816 / 4096
Monitoring (mon_tid)	6 (Low)	0%	864 / 4096
System Idle	N/A	48%	208 / 1024
Mutex Block Time	N/A	0 ms	N/A

- Scheduling and CPU Utilization
- The Zephyr kernel demonstrated highly deterministic task scheduling and efficient resource allocation. The high-priority measurement thread, tasked with the time-critical 1kHz ADC polling loop, consumed an average of 37% of the CPU cycles. Despite this intensive continuous polling, the system maintained a robust 48% idle state. This significant idle headroom indicates that the Zephyr implementation is computationally efficient, easily accommodating the high-frequency sampling rate without approaching the microcontroller's processing limits. Furthermore, the LCD display thread registered a nominal 0% CPU load, confirming that Zephyr's hardware abstraction layer and I2C drivers handle blocking peripheral communications seamlessly, preventing user-interface updates from starving the critical sensing loops.
- Concurrency and Memory Management
- Resource contention was rigorously evaluated by monitoring the access latency to the shared global variables via the `data_mutex`. The Zephyr implementation consistently recorded a 0 ms block time. This proves that the producer-consumer architecture executed flawlessly; the high-priority measurement task was never forced to wait on the lower-priority calculation or display tasks, completely avoiding priority inversion.
- Memory management in Zephyr, which relies on compile-time static allocation, proved highly predictable. The monitoring thread utilized the largest stack space at 864 bytes, while the intensive measurement thread required only 464 bytes. While this static allocation strategy requires a larger overall memory footprint upfront compared to dynamic allocation, it effectively eliminates the risk of runtime heap fragmentation—a critical advantage for the long-term reliability of industrial energy meters. While minor UART buffer overruns (noted as dropped messages in the system logs) were observed during serial telemetry output, these were strictly isolated to the logging subsystem and did not impede the real-time data acquisition or RMS calculations.
- Next Steps for your report: To finish the comparison, you just need to capture the same metrics (Mutex Time, CPU Load, Stack size) from your FreeRTOS version. Once you have those numbers, you can add a final paragraph comparing the two directly!
- The deployment of the FreeRTOS firmware on the ESP32 successfully validated both the physical hardware integration and the efficiency of the dynamic RTOS scheduler. Unlike the initial configuration hurdles encountered with the Zephyr driver tree, the FreeRTOS implementation (built on the Arduino core) immediately established communication with the ADC channels and the I2C display. The system was tested under the same steady-state 230V AC mains load using the 9W LED bulb, capturing accurate real-time electrical metrics alongside system performance telemetry.

• **Table 2: FreeRTOS Performance and Telemetry Metrics**

Metric / Task	Priority	State	Observed Value / Details
Voltage (RMS)	N/A	N/A	~237.5 V
Current (RMS)	N/A	N/A	~0.03 A
Active Power	N/A	N/A	~8.0 W
MeasTask	3 (High)	Blocked (2)	High-frequency sampling loop

Metric / Task	Priority	State	Observed Value / Details
CalcTask	2	Blocked (2)	RMS math processing
LCDTask	1	Ready (1)	I2C display rendering
MonitorTask	1 (Low)	Running (0)	System telemetry output
Free Heap Memory	N/A	N/A	254,496 Bytes
Mutex Block Time	N/A	N/A	0 ms

- Hardware Integration and Measurement Accuracy
- The serial output confirms the system is correctly acquiring and interpreting analog signals. The calculated values (approximately 237V and 8W) align closely with the expected parameters of the regional grid and the physical 9W LED load. The minor deviation in active power (recording slightly lower than the rated 9W) is a standard characteristic of capacitive LED power supplies, validating that the firmware's root-mean-square math and offset calibration (calibrateOffsets) are functioning accurately in a real-world electrical environment.
- Task Concurrency and Synchronization
- The FreeRTOS scheduler demonstrated exceptional efficiency in managing the dual-core ESP32 architecture. Task synchronization was handled via a SemaphoreHandle_t mutex (dataMutex), which protects the global variables during the data transition between the high-speed measurement task and the slower calculation/display tasks. Telemetry data reveals a constant **Mutex Block Time of 0 ms**. This indicates zero priority inversion and zero contention bottlenecks; the MeasTask is able to lock, update, and unlock the data structures instantaneously without being stalled by the lower-priority threads.
- Memory Footprint and State Management
- In contrast to Zephyr's static compile-time allocation, FreeRTOS utilizes dynamic memory allocation on the heap. The esp_get_free_heap_size() metric remained stable at 254,496 bytes throughout the execution. This proves that the system's memory footprint is highly optimized and free of memory leaks, leaving over 250KB of RAM available for potential future expansions (such as Wi-Fi telemetry or MQTT streaming). Furthermore, the task state monitoring (uxTaskGetSystemState) confirms that tasks are appropriately yielding execution time; MeasTask and CalcTask rest in a Blocked state (State 2) using vTaskDelay when not actively polling, preventing CPU exhaustion and allowing the IDLE tasks to execute.

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